RODIA

Rocket Onboard Drone Interface Avionics

Project brief

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1 Introduction

This document briefly describes the development of a flight computer for Trinity, a rocket built by Manhattan Rockets, along with the motivation behind it, its goals and its requirements. It is written by Vít Krejčí, a member of the avionics team of Manhattan Rockets, a student amateur rocketry team operating under the Czech Rocket Society.

2 Motivation

In our rocket team, Manhattan Rockets, we are developing a hybrid vehicle that combines amateur rocketry with modern FPV drone technology. Trinity's operating profile spans two fundamentally different regimes: a ballistic rocket-phase ascent, and a controlled, powered drone-phase descent. The drone phase, including self-stabilisation of the rocket-drone structure and the controlled descent, is handled mainly by a drone flight controller.

The motivation for adding a second flight computer comes from the limitations of drone flight controllers in this role. Their firmware provides only filtered data, tuned for propeller-driven flight rather than for a rocket ascent. They are also not built for rocket conditions, and the accelerometer is a particular limitation: the g-forces of our flight come close to the upper measurement range of typical drone-grade sensors. Running two flight computers built on different principles also reduces the risk of a single failure ending the flight, since they do not share the same firmware or the same weaknesses.

We also want the ability to write our own firmware, which is not realistically possible when using drone flight controllers alone. A further reason for building a new board is to learn electrical engineering in this specific context.

3 Idea

The idea is to build our own board that connects easily to a drone flight controller and its peripherals, and adds the capabilities that drone flight controllers do not have. The board operates independently, so that if the drone flight controller fails, everything critical can still be carried out.

At its core this is a division of responsibility between two units. The drone controlling unit handles what it is already good at; the custom board handles what the drone system was never designed for, and is built to operate reliably under rocket-flight conditions.

The approach could extend beyond this project. Because the board is easily programmable and built to the mechanical and electrical standards of drone flight controllers, anyone could connect standard drone peripherals to it, such as GPS, cameras, or antennas, rather than developing a complete avionics stack from scratch.

4 Requirements

- The board must be able to carry out the most critical functions of the rocket on its own, including parachute deployment via servo or pyrotechnic e-match.
- It must collect data from its own clock, IMU, accelerometer, barometer and temperature sensor, store it to an SD card, and transmit it to the ground station.
- It must communicate with the drone flight controller.
- It must operate reliably under rocket-flight conditions.
- It must be built around an RP-series microcontroller, as in our previous iterations.
- It must be easily programmable, which follows from the choice of microcontroller above.
- Its accelerometer must cover the g-forces expected in flight, with adequate margin.
- It must have its own power circuit, since it will run from a separate battery to the drone flight controller.
- It must include a circuit for a pyro channel capable of fireing an e-match.
- It must operate across an input voltage range of at least 5 V to 8.4 V, covering supply from 5 V sources as well as a fully charged 2S battery.
- The accelerometer and barometer must provide fast and precise enough measurements for them to be idealy better than the ones from drone FC or being comparable to them. As already mentioned they will provide the system with unfiltered data.
- The SD card or as called black box must be able to handle the sample rate and size of logging from all sensors.
- It must read its battery level.
- It must follow the physical standards of drone flight controllers, including board dimensions, mounting hole pattern, and the type and placement of connectors.
- The board must be able to handle the current of everything onboard plus multiple servo motors.

5 Scheme

The diagram below is a heavily simplified representation of the system. It shows only the main functional blocks and how they relate to one another; individual components, connections and the internal structure of each unit are omitted. The arrows indicate the direction of data flow.

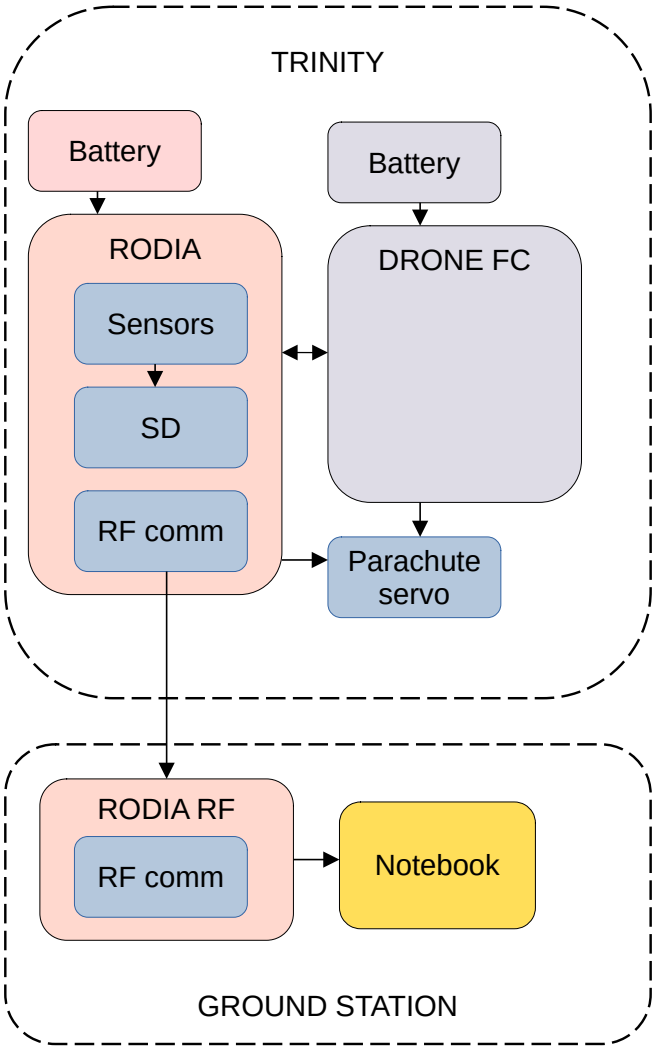


Figure 1: Simplified block diagram of the RODIA and drone flight controller architecture